

Lab 04: Router Configuration & Subnetting

Subnetting a Network and Configuring Static Routes

Lab #4

Topic: Subnetting &
Routing

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Divide a given IP block into multiple subnets using CIDR notation.
- Configure router interfaces with correct subnet IPs and masks.
- Apply static routes so traffic can cross between subnets.
- Verify inter-subnet connectivity using ping.
- Relate to Network+ objective: 1.4 – Explain the characteristics of routing concepts.

2. Network Topology

The base network 192.168.1.0/24 is divided into two /25 subnets. Router R1 has two FastEthernet interfaces, each facing one subnet. PC0 and PC1 sit on opposite subnets and communicate via R1.

Device	Interface	IP Address	Subnet	Subnet Mask
R1	Fa0/0	192.168.1.1	192.168.1.0/25	255.255.255.128
R1	Fa0/1	192.168.1.129	192.168.1.128/25	255.255.255.128
PC0	Fa0	192.168.1.10	192.168.1.0/25	255.255.255.128
PC1	Fa0	192.168.1.130	192.168.1.128/25	255.255.255.128

3. Subnetting Reference

Dividing 192.168.1.0/24 into two equal /25 subnets:

Subnet	Network Address	First Host	Last Host	Broadcast	Hosts
Subnet 1	192.168.1.0	192.168.1.1	192.168.1.126	192.168.1.127	126
Subnet 2	192.168.1.128	192.168.1.129	192.168.1.254	192.168.1.255	126

4. Step-by-Step Configuration

4.1 Router R1 – Both Interfaces

```
Router> enable
Router# configure terminal
Router(config)# hostname R1

R1(config)# interface FastEthernet0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.128
R1(config-if)# no shutdown
R1(config-if)# exit

R1(config)# interface FastEthernet0/1
R1(config-if)# ip address 192.168.1.129 255.255.255.128
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# exit
R1# write memory
```

4.2 PC IP Configuration

PC0 – IP	192.168.1.10
PC0 – Mask	255.255.255.128
PC0 – GW	192.168.1.1
PC1 – IP	192.168.1.130
PC1 – Mask	255.255.255.128
PC1 – GW	192.168.1.129

5. Verification

```
PC0> ping 192.168.1.130    ! Cross-subnet ping via R1
PC0> ping 192.168.1.1      ! Ping own gateway
R1# show ip interface brief  ! Verify interfaces are up/up
R1# show ip route           ! Confirm connected routes
```

6. Key Takeaways

- A /25 mask (255.255.255.128) splits a /24 into two subnets of 126 usable hosts each.
- Routers route between subnets — without a router, hosts on different subnets cannot communicate.
- Connected routes are automatically added to the routing table when an interface is configured and up.

- Always verify interfaces are up/up with **show ip interface brief** before troubleshooting routing.
- The 'write memory' command saves the configuration to NVRAM so it survives reboots.